Batch Effect Diagnostics for Taxonomy-aware BiomeGPT

Yuanshu / Codex working report

May 20, 2026

Executive Summary

We evaluated whether the taxonomy-aware BiomeGPT phase-2 sample embeddings encode batch/study information and whether scGPT-style adversarial batch correction is a reasonable next direction. The short answer is:

- The batch labels used here are not raw metadata columns. They come from an externally enriched study/cohort batch-proxy annotation built from public accession lookup and conservative local-prefix rules.
- After receiving the uploaded real study-id table `BiomeGPT_species_samples_studyIDs.csv`, we reimplemented the batch diagnostics and correction trials using true sample-to-study alignment rather than proxy labels.
- The phase-2 sample embeddings do contain measurable batch signal.
- However, most broad batch labels are strongly phenotype-confounded, so using all labels for adversarial correction would risk removing true disease biology.
- A conservative proof-of-concept using only safer batch labels ran end-to-end and reduced safe-batch predictability modestly while mostly preserving Healthy-vs-Diseased signal.
- This is a working diagnostic and correction pipeline, but not yet a final batch-corrected model.

The most important numerical result is shown below:

Probe	Before macro-F1	After macro-F1	Before bal. acc.	After bal. acc.
Safe batch labels	0.499	0.476	0.503	0.475
Healthy vs Diseased	0.796	0.789	0.801	0.794

This means the light adversarial correction moved in the desired direction: batch predictability decreased, while the biological Healthy-vs-Diseased signal was largely preserved.

The newer real-study-id experiments are more stringent and should now be treated as the main evidence. With true study ids, study signal is strong, phenotype confounding is substantial, and naive GRL adversarial fine-tuning did not yet remove study signal. A post-hoc representation correction can remove study signal almost completely, but broad study correction also removes much of the H/D signal, confirming that study and phenotype are entangled in this dataset.

1 Scientific Question

After building the taxonomy-aware BiomeGPT model, the next question was whether the learned sample-level representations contain unwanted technical or cohort-specific batch information. This matters because external validation performance may be limited by study/cohort shift, and scGPT-style batch integration suggests a possible solution: make sample embeddings less predictive of batch while retaining biological phenotype information.

The goal of this stage was therefore not to immediately claim batch correction, but to answer three diagnostic questions:

1. Can batch/study labels be predicted from the BiomeGPT sample prompt?
2. Are those batch labels safe to remove, or are they confounded with phenotype?
3. If we run a conservative adversarial correction, does batch predictability go down without destroying Healthy-vs-Diseased signal?

2 Batch Annotation Construction

2.1 Why the data had to be processed

The original phase-2 gut metadata did not contain an explicit technical batch column. It also did not include a clean study ID, cohort ID, sequencing center, sequencing platform, country, library preparation, or run-date column. The available fields were:

`sample_id`, `Phenotype`, `body_site`, `Phenotype_fullname`

Because adversarial batch correction requires a batch or domain label, we first had to reconstruct the best available batch proxy. The goal was not to claim that every technical sequencing batch is known. Instead, the goal was to build an auditable study/cohort-level proxy that is good enough for diagnostic probing and cautious enough for conservative correction experiments.

2.2 How sample IDs were interpreted

The key observation was that many `sample_id` values contain public database accessions. These accessions can be queried against public metadata services to recover study or cohort information. Each sample ID was first classified by pattern:

Pattern	Interpreted type
`SAMN...`	NCBI BioSample accession
`SAMEA...`	ENA/EBI BioSample accession
`SAMD...`	DDBJ/INSDC BioSample accession
`SRR...`	NCBI SRA run accession
`EGAR...`	EGA run accession
Other	Local or unknown sample ID

For compound IDs such as `EGAR00001420100_9002000001328080LL`, only the public run accession part was used for external lookup, while the original full sample ID was preserved.

2.3 External lookup strategy

Public accession-like IDs were mapped to study or cohort metadata using official public services:

Source	What it provided
ENA Portal API	Sample-to-study metadata for `SAMN`, `SAMEA`, and `SAMD` IDs
NCBI SRA RunInfo	Run-to-BioProject/SRAStudy metadata for `SRR` IDs
NCBI BioSample E-utilities	Fallback for older `SAMN` records missing usable ENA study accessions
EGA Metadata API	Run-to-study metadata for `EGAR` IDs

The recommended batch label was then constructed from the strongest available study or cohort identifier. Examples include:

`study:PRJNA763023`, `study:PRJEB103248+PRJEB22893`, `dbgap:phs000228`, `ega_study:EGAS00001001704`

If a sample mapped to multiple study accessions, the accessions were joined with `+`. This can happen when a public database links both an original project and a derived metagenomic assembly project.

2.4 Local IDs and confidence levels

Local sample IDs without public accessions could not be externally verified from this metadata alone. They were therefore handled cautiously. A prefix-derived label was assigned only when the local naming pattern was structured and biologically interpretable. Examples include `PRIMM...`, `M...`, `SID...`, `wHAXPI...`, and `FMT...`. These labels are useful for diagnostics and manual review, but they are not treated as fully verified external batch labels.

Every sample received an external confidence label:

Confidence	Samples	Meaning
High	9,389	Exact public accession lookup returned study/cohort metadata
Medium	3,418	Structured local prefix-derived cohort label
Low	525	Unresolved or ambiguous local sample ID pattern

This distinction matters because the batch probe can use broad labels for exploratory diagnostics, but final correction should not treat prefix-derived or unresolved labels as equally reliable.

2.5 Phenotype-confounding check

A real study label is not automatically safe for batch correction. Many microbiome studies are highly phenotype-specific. For example, one public study may contain only Healthy controls, while another may contain only Parkinson's disease samples. If we train an adversarial model to remove those study labels, the model may also remove true disease biology.

Therefore, each proposed batch label was checked for phenotype composition:

`n_samples`, `n_phenotypes`, `top_phenotype`, `top_phenotype_count`, `top_phenotype_fraction`

The warning rule was:

`phenotype_confounding_warning=True` if `top_phenotype_fraction >= 0.80` or `n_phenotypes <= 1`

This is intentionally strict. If a batch group is nearly one phenotype, correcting away that batch may also correct away the phenotype signal we want the model to preserve.

2.6 Conservative-safe label rule

For final batch-correction experiments, a sample was marked:

`safe_for_final_batch_correction_conservative=True`

only if both conditions were satisfied:

`external_confidence == "high"` and `phenotype_confounding_warning == False`

This produced:

Category	Samples
Conservative-safe for final batch correction	2,134
Not conservative-safe	11,198

The resulting annotation should therefore be described as an externally enriched study/cohort batch-proxy annotation, not as a perfect technical sequencing-batch table. The broad `batch_label_external_recomme` field is appropriate for asking whether embeddings contain study/cohort signal. The conservative-safe subset is the appropriate starting point for asking whether adversarial correction can reduce batch signal without destroying phenotype signal.

3 Inputs Used

3.1 Model checkpoint

We used the taxonomy-aware phase-2 checkpoint:

`dataset_v3/outputs_taxonomy_notebook/taxonomy_checkpoint_stage2.pt`

This checkpoint is the full taxonomy-aware run, not the earlier minimal BioGPT checkpoint. Its model structure contains rank-wise taxonomy embeddings, including Domain, Kingdom, Phylum, Class, Order, Family, Genus, and Species embeddings.

3.2 Data

The diagnostic data came from the phase-2 gut pretraining set and the externally enriched batch annotation file:

`dataset_v3/abund_pretraining_phase2_gut.csv.zip`

`dataset_v3/meta_pretraining_phase2_gut_batch_annotation_external_enriched.csv`

The final diagnostic matrix contained:

Quantity	Count
Phase-2 gut samples	13,332
Sample embedding dimension	512
Checkpoint species vocabulary	1,012
Unique recommended batch labels	218
High-confidence external labels	9,389
Medium-confidence labels	3,418
Low-confidence labels	525
Conservative-safe batch-correction samples	2,134

4 Method: Sample Prompt Extraction

For each phase-2 gut sample, we extracted the BiomeGPT sample prompt, defined as the final-layer `<cls>` embedding from the pretrained taxonomy-aware transformer. The extraction procedure was:

1. Load the taxonomy-aware phase-2 checkpoint.
2. Load the phase-2 abundance matrix.
3. Align abundance rows to the batch annotation metadata by sample ID.
4. Reindex abundance columns to the checkpoint species vocabulary.
5. Convert relative abundance values to 32 rank-based abundance bins.
6. Run the pretrained model in inference mode and save the `<cls>` embedding.

This produced an embedding matrix:

```
X_sample ∈ R^(13332 x 512)
```

The cached output is:

```
dataset_v3/outputs_batch_diagnostics_taxonomy/phase2_sample_prompt_embeddings.npz
```

5 Method: Batch Probe

5.1 What is a batch probe?

A batch probe is a shallow supervised classifier trained on frozen model embeddings. Here, the input is the sample prompt `<cls>` embedding, and the target is a batch/study label. If the classifier predicts batch labels much better than a dummy baseline, then the embedding contains batch information.

Importantly, this does not modify the BiomeGPT model. It is a diagnostic measurement.

5.2 Probe model

For each probe, we used:

- Standard scaling of the 512-dimensional embeddings.
- Multiclass logistic regression.
- Class-balanced weighting.
- A stratified 80/20 train-test split when feasible.
- Metrics: accuracy, balanced accuracy, and macro-F1.
- Dummy baseline: most-frequent-class classifier.

Balanced accuracy and macro-F1 are more informative than raw accuracy because batch and phenotype labels are imbalanced.

5.3 Batch label panels

We evaluated three batch-label panels:

1. **High + medium labels**: broadest usable set for hypothesis generation.
2. **High-only labels**: exact external accession lookups where available.
3. **Conservative-safe labels**: labels passing the conservative safety rule for final batch correction.

For each panel, labels with fewer than 10 samples were excluded from the probe to avoid unstable tiny-class estimates.

6 Batch Probe Results

Probe target	Samples	Classes	Bal. acc.	Macro-F1	Dummy bal. acc.
High + medium batch labels	12,758	105	0.397	0.365	0.010
High-only batch labels	9,340	83	0.386	0.355	0.012
Conservative-safe batch labels	2,134	17	0.503	0.499	0.059
Phenotype labels	13,332	30	0.547	0.395	0.033
Healthy vs Diseased	13,332	2	0.801	0.796	0.500

6.1 Interpretation

The batch signal is real. For conservative-safe labels, the batch probe reached macro-F1 0.499, compared with dummy macro-F1 0.014. This is much higher than chance and shows that study/batch information is encoded in the taxonomy-aware phase-2 sample prompts.

At the same time, the embeddings also retain strong biological signal: a shallow Healthy-vs-Diseased probe reached macro-F1 0.796. This is important because batch correction should reduce batch signal without removing phenotype signal.

7 Method: Batch-Phenotype Confounding Analysis

Before doing any correction, we checked whether batch labels were confounded with phenotype. This is critical because if a batch label is nearly identical to a disease label, then adversarially removing batch information may also remove real disease biology.

For each batch-label panel, we computed:

- Batch-by-phenotype contingency table.
- Cramer's V between batch and phenotype.
- Normalized mutual information (NMI).
- Top phenotype fraction per batch.
- Number of batches where the top phenotype fraction is at least 0.8.

The top phenotype fraction is especially interpretable. For example, if one batch contains 100 samples and 95 are Healthy, its top phenotype fraction is 0.95. Correcting away that batch could also correct away Healthy-specific biology.

8 Confounding Results

Label set	Cramer's V	NMI	Median top phenotype frac.	Batches top frac. ≥ 0.8
High + medium	0.802	0.519	1.000	79 / 105
High-only	0.907	0.489	1.000	66 / 83
Conservative-safe	0.683	0.457	0.549	0 / 17

8.1 Interpretation

The broad high+medium and high-only batch labels are strongly phenotype-confounded. Their median top phenotype fraction is 1.000, meaning many batches contain only one phenotype. These labels are not safe for direct adversarial removal.

The conservative-safe label set is not perfectly independent of phenotype, but it is much less confounded. Its median top phenotype fraction is 0.549, and none of its 17 retained batches has top phenotype fraction above 0.8. Therefore, this set is the best starting point for a cautious batch-correction proof-of-concept.

9 Conservative Adversarial Correction Trial

9.1 Why adversarial correction?

scGPT-style batch integration uses a domain-adversarial branch: a discriminator tries to predict batch from the sample representation, while gradient reversal makes the encoder learn representations that are less batch-predictive. We adapted that idea to BiomeGPT by using the `<cls>` sample prompt as the representation for batch discrimination.

9.2 What exactly was corrected?

This was not a new pretraining run from scratch. The trial continued training from the existing taxonomy-aware phase-2 checkpoint:

`dataset_v3/outputs_taxonomy_notebook/taxonomy_checkpoint_stage2.pt`

The correction target was the sample-level `<cls>` embedding produced by the taxonomy-aware transformer. This is the same sample prompt used by the downstream probes. The goal was to make this `<cls>` embedding less predictive of conservative-safe study/cohort labels while keeping the original masked-abundance reconstruction behavior.

We did not train on all batch labels. The adversarial branch used only rows satisfying:

`safe_for_final_batch_correction_conservative=True`

and retained only batch classes with at least 10 samples. This gave:

Quantity	Value
Safe training samples	2,134
Batch classes	17
Minimum retained class size	43
Maximum retained class size	278

9.3 Implementation details

The implementation is in:

`dataset_v3/batch_adversarial_correction.py`

The data flow was:

1. Load the taxonomy-aware phase-2 checkpoint.
2. Load the phase-2 gut abundance matrix.
3. Align abundance rows to the enriched batch annotation table.
4. Filter to conservative-safe samples.
5. Reindex abundance columns to the checkpoint species vocabulary.
6. Convert abundance values to 32 rank-based bins.
7. Continue training the BiomeGPT encoder with the original reconstruction objective plus a small adversarial batch objective on the `<cls>` token.

9.4 Batch discriminator

The adversarial branch was a small multilayer perceptron attached to the `<cls>` embedding. Its architecture was:

```
LayerNorm(512) -> Linear(512, 256) -> ReLU -> Dropout(0.1)
-> Linear(256, 128) -> ReLU -> Dropout(0.1) -> Linear(128, 17)
```

The discriminator's direct task is ordinary multiclass batch classification. It receives the `<cls>` embedding and predicts one of the 17 conservative-safe batch labels.

9.5 Gradient reversal mechanism

The important point is that the discriminator is connected through a gradient reversal layer (GRL). In the forward pass, GRL behaves like the identity function:

```
GRL(x) = x
```

In the backward pass, it multiplies the gradient by `-lambda`. In this trial:

```
grl_lambda = 1.0
```

This creates an adversarial objective:

- The discriminator learns to predict batch from the `<cls>` embedding.
- The encoder receives the reversed gradient, so it learns to make the `<cls>` embedding less useful for batch prediction.

Therefore, the discriminator and encoder have opposite goals. The discriminator tries to recover batch labels; the encoder tries to hide batch information while still solving the masked abundance reconstruction task.

9.6 Training objective

For each mini-batch, we generated a random species-token mask using:

```
mask_ratio = 0.25
```

The model then encoded the binned abundance vector with the mask applied. The species token hidden states were sent to the reconstruction head, and the `<cls>` hidden state was sent to the batch discriminator.

The optimized loss was:

```
total loss = reconstruction loss + dab_weight * batch cross-entropy loss
```

with:

```
dab_weight = 0.1
```

The reconstruction loss was mean-squared error on masked nonzero abundance bins. The batch loss was cross-entropy over 17 conservative-safe batch classes. Because the batch loss passes through GRL, minimizing this total loss trains the discriminator normally but pushes the encoder in the opposite direction for the batch component.

9.7 Why this is a light correction

This run was intentionally conservative. The settings were:

Hyperparameter	Value
Epochs	3
Training batch size	8
Learning rate	2e-5
Weight decay	1e-4
Mask ratio	0.25
DAB weight	0.1
GRL lambda	1.0
Discriminator dropout	0.1

It is "light" because the model is only continued for 3 epochs and the adversarial term has small weight. The main training pressure remains the original masked-abundance reconstruction objective. This reduces the risk of damaging the pretrained microbiome representation before we have downstream validation.

9.8 Checkpoint and evaluation workflow

After the 3-epoch correction run, the corrected model was saved as:

```
dataset_v3/outputs_batch_adversarial_taxonomy/taxonomy_checkpoint_stage2_batch
_adversarial_safe.pt
```

Then we re-extracted `<cls>` embeddings for all 13,332 phase-2 gut samples from the corrected checkpoint and reran the same frozen logistic-regression probes:

- safe-batch probe on conservative-safe batch labels;
- Healthy-vs-Diseased probe on all samples.

The desired outcome was not maximum downstream disease performance yet. The immediate criterion was directional: safe-batch probe performance should decrease, while Healthy-vs-Diseased probe performance should remain close to the uncorrected baseline.

10 Adversarial Trial Results

10.1 Trial A: light correction

Settings:

- Epochs: 3
- DAB weight: 0.1
- Gradient reversal lambda: 1.0
- Safe training samples: 2,134

- Batch classes: 17

Training history:

Epoch	Total loss	Reconstruction loss	DAB loss	DAB train acc.
1	37.924	37.645	2.790	0.103
2	37.471	37.198	2.733	0.143
3	37.607	37.337	2.703	0.147

The DAB training accuracy increased from 0.103 to 0.147, showing that the discriminator was learning some batch signal during training. Because the discriminator was connected through GRL, that same signal produced reversed gradients for the encoder.

Before/after probe results:

Probe	Before macro-F1	After macro-F1	Before bal. acc.	After bal. acc.
Safe batch labels	0.499	0.476	0.503	0.475
Healthy vs Diseased	0.796	0.789	0.801	0.794

This is the best current trial. Batch predictability decreased modestly, while Healthy-vs-Diseased predictability was mostly preserved.

10.2 Trial B: stronger correction

Settings:

- Epochs: 5
- DAB weight: 0.5
- Gradient reversal lambda: 1.0

Probe	Before macro-F1	After macro-F1	Before bal. acc.	After bal. acc.
Safe batch labels	0.499	0.505	0.503	0.505
Healthy vs Diseased	0.796	0.786	0.801	0.790

This stronger run did not improve batch correction. Batch predictability slightly increased, and H/D signal decreased slightly. This suggests that simply increasing adversarial weight is not sufficient; the schedule and training design need tuning.

11 Additional Label-Panel Ablation

After the initial conservative-safe run, we tested three label-panel choices under the same light adversarial setting:

1. **High + medium labels**: all exact external labels plus structured prefix-derived labels.
2. **High-only labels**: exact external database labels only, without removing phenotype-confounded batches.

3. **Conservative-safe labels**: exact external labels after removing strongly phenotype-confounded batches.

The purpose was to separate two questions:

- Does using broader labels create stronger batch removal?
- Does increasing training scale or strength necessarily improve correction?

For this ablation, the script was generalized with:

```
--label_panel high_medium | high_only | conservative_safe
```

and the three runs used the same light correction hyperparameters:

```
epochs=3, dab_weight=0.1, grl_lambda=1.0, train_batch_size=32
```

11.1 Ablation results

Training panel	Samples	Classes	Batch macro-F1 before	Batch macro-F1 after
High + medium	12,758	105	0.365	0.383
High-only	9,340	83	0.355	0.373
Conservative-safe	2,134	17	0.499	0.511

Training panel	Batch bal. acc. before	Batch bal. acc. after	H/D macro-F1 after
High + medium	0.397	0.411	0.804
High-only	0.386	0.405	0.802
Conservative-safe	0.503	0.513	0.794

In this batch-size-32 ablation, none of the three label-panel runs reduced the target batch probe. The high+medium and high-only runs actually increased batch predictability. This is not the desired correction direction.

11.2 Comparison against earlier runs

The best directional result is still the earlier conservative-safe light run with `train_batch_size=8`:

Run	Batch macro-F1 before	Batch macro-F1 after	H/D ma
Conservative-safe, batch size 8, DAB 0.1, 3 epochs	0.499	0.476	
Conservative-safe, batch size 32, DAB 0.1, 3 epochs	0.499	0.511	
Conservative-safe, DAB 0.5, 5 epochs	0.499	0.505	

This answers the "should we simply train more or larger?" question: probably not by itself. Stronger or larger updates did not reliably improve batch removal. In fact, the broader label panels and the stronger DAB run moved in the wrong direction for the batch probe.

11.3 Interpretation

These ablations support three conclusions:

- Broad high+medium labels are useful for diagnostics, but they are too heterogeneous and phenotype-confounded for naive adversarial correction.
- High-only labels are externally verified, but many are still phenotype-confounded; external verification alone does not make a label safe for correction.
- Conservative-safe labels remain the most scientifically defensible correction target, but optimization is sensitive. We should tune schedule and validation criteria rather than only increasing epochs or DAB weight.

The next technical step should be a small hyperparameter sweep on conservative-safe labels: for example, DAB weight `0.02, 0.05, 0.1`, batch size `8 or 16`, and a gradual GRL/DAB warm-up schedule. The selection criterion should require lower conservative-safe batch probe performance while preserving Healthy-vs-Diseased probe performance.

12 Masked Abundance Reconstruction Diagnostic

12.1 Why this diagnostic matters

BiomeGPT is a foundation model whose pretraining objective is masked abundance prediction. Healthy-vs-Diseased classification is a downstream task. Therefore, batch correction should not be evaluated only by H/D probe performance. A better foundation-model criterion is:

> Does correction reduce batch predictability while preserving or improving masked abundance reconstruction?

To evaluate this directly, we added:

`dataset_v3/batch_reconstruction_diagnostics.py`

For each checkpoint, the script applies the same deterministic mask to nonzero abundance bins and computes reconstruction MSE on the masked positions. It also computes batch-stratified MSE summaries for the selected label panels.

12.2 Initial reconstruction results

Checkpoint	Overall MSE	Safe-panel MSE	Safe-panel CV	Safe-panel gap
Original phase-2	34.857	37.808	0.095	13.140
Conservative-safe, batch size 8	33.745	35.724	0.093	12.961
Conservative-safe, batch size 32	34.088	36.252	0.096	13.324
High + medium, batch size 32	33.049	35.770	0.096	12.736
High-only, batch size 32	33.373	36.061	0.098	13.453
Conservative-safe, DAB 0.5	33.853	35.256	0.092	12.554

The main observation is that continued correction training did not destroy the masked abundance objective. Overall reconstruction MSE decreased for all corrected checkpoints relative to the original phase-2 checkpoint. The best conservative-safe batch-size-8 run also reduced safe-panel reconstruction MSE and slightly reduced the safe-panel batch-MSE gap.

12.3 Interpretation

This improves the evaluation design. The correct selection criterion should have three layers:

1. **Batch removal**: post-hoc batch probe should decrease.
2. **Foundation objective preservation**: masked abundance reconstruction MSE should remain stable or improve, ideally with lower per-batch MSE gap/CV.
3. **Downstream biological sanity**: H/D probe or ExVal performance should not collapse.

Importantly, lower reconstruction MSE alone is not enough. Some runs improved reconstruction while the batch probe increased. This means abundance modeling and batch removal must be evaluated jointly.

13 Discussion and Scientific Insights

13.1 Batch labels are real, but not all are safe correction targets

The batch probe results show that the taxonomy-aware BiomeGPT sample embeddings encode study/cohort-associated signal. This supports the motivation for batch-effect diagnostics. However, the confounding analysis shows that many candidate batch labels are still tied to phenotype. This is especially true for the broad high+medium and high-only panels.

The main insight is:

> External study labels are useful for diagnostics, but external verification alone does not make them safe for correction.

A label can be a real public study label and still be unsafe for adversarial correction if the study contains mostly one disease phenotype.

13.2 Phenotype confounding likely explains broad-label failure

For both broad-label adversarial trials, the target batch probe increased after training:

Panel	Batch macro-F1 before	Batch macro-F1 after
High + medium	0.365	0.383
High-only	0.355	0.373

This suggests that the model did not learn a clean technical-batch removal operation. Because study and phenotype are entangled, the adversarial objective may push the encoder into a new representation that still separates study/cohort structure, or even separates phenotype-associated study structure more clearly.

13.3 More aggressive training is not automatically better

The stronger conservative-safe run also did not improve batch removal:

Run	Batch macro-F1 after	H/D macro-F1 after
Conservative-safe, DAB 0.1, batch size 8, 3 epochs	0.476	0.789
Conservative-safe, DAB 0.5, 5 epochs	0.505	0.786

Therefore, the current limitation is probably not simply that the model was trained too little. Increasing DAB weight or epochs can make the adversarial objective less stable and may damage biological signal without removing batch signal.

13.4 The discriminator-encoder game may be unstable

Adversarial correction is a two-player optimization problem. The discriminator tries to predict batch from `<cls>`; the encoder tries to hide batch from the discriminator; the reconstruction loss tries to preserve microbiome abundance structure. If the discriminator learns too quickly, too slowly, or from noisy/confounded labels, the reversed gradient may not point toward a clean batch-invariant representation.

This instability is a plausible explanation for why several runs increased post-hoc batch probe performance instead of decreasing it.

13.5 The encoder may learn new batch-separable structure

The adversarial discriminator only penalizes the batch directions it can exploit during training. After correction, we evaluate with a fresh frozen logistic-regression probe. If the after-probe performs better, that means batch information is still present, possibly in a different geometry. In other words, the encoder may hide batch from the training discriminator while still producing embeddings that remain separable by another classifier.

This is why post-hoc probes are important: they test whether the final embedding space is actually less batch-predictive, not merely whether the training discriminator struggled.

13.6 No warm-up schedule is a weakness

The current implementation applies fixed adversarial pressure from the beginning:

```
grl_lambda = 1.0,   dab_weight = fixed
```

This may be too abrupt. A more stable domain-adversarial setup would warm up the adversarial signal gradually:

> start with low or zero DAB pressure, let reconstruction stabilize, then gradually increase GRL/DAB strength.

This could prevent early adversarial gradients from pushing the pretrained representation into an unstable or newly batch-separable geometry.

13.7 Batch size changes adversarial dynamics

Batch size is not only a speed parameter in adversarial training. The best directional result used conservative-safe labels with batch size 8:

```
batch macro-F1:  0.499 -> 0.476
```

but the otherwise similar batch-size-32 conservative-safe run moved in the wrong direction:

```
batch macro-F1:  0.499 -> 0.511
```

This suggests that minibatch composition affects the discriminator-encoder game. Smaller batches may regularize the adversarial dynamics differently, while larger batches may make the discriminator updates smoother but not necessarily more useful for batch-invariant representations.

13.8 Overall interpretation

The data support the batch-correction idea, but not a final corrected model yet. A concise interpretation is:

> Taxonomy-aware BiomeGPT embeddings encode study/cohort-associated signal. However, broad batch labels are strongly phenotype-confounded, and naive adversarial correction can fail or even increase post-hoc batch separability. The most promising direction is conservative-safe labels with a more careful adversarial schedule, rather than simply using more labels, more epochs, or a stronger DAB loss.

14 Real Study-ID Reimplementation

14.1 Why this was necessary

The first batch pipeline used externally enriched proxy batch labels because the raw phase-2 metadata did not directly expose a clean study-id field. After the file `BiomeGPT_species_samples_studyIDs.csv` became available, we rebuilt the batch annotation around true sample-to-study mapping. This is scientifically better because the batch variable is now a real cohort/study identifier rather than an inferred proxy.

The uploaded CSV had sample ids in the first column and study names in `study_name`. Some rows contained unmatched quote characters, so the parser reads the file with quoting disabled, then strips quotes and normalizes whitespace. We then align by exact `sample_id` to `dataset_v3/meta_pretraining_phase2_gut.csv`.

14.2 Real study-id coverage

Quantity	Count
Rows in uploaded study-id file	13,524
Phase-2 gut samples	13,332
Phase-2 samples matched to real study id	13,326
Phase-2 samples missing real study id	6
Unique real studies in phase-2 gut	80
Conservative-safe samples	4,751
Conservative-safe studies	26

For matched samples, the new batch label is:

```
batch_label_external_recommended = real_study:<study_name>
```

All matched real-study labels are treated as high-confidence because they come from the uploaded explicit sample-to-study table. The old proxy labels remain useful history, but the real-study results supersede them for the next round of experiments.

14.3 Real study-id diagnostic probes

Using the original taxonomy-aware phase-2 checkpoint, we re-extracted the 512-dimensional sample-prompt embeddings and trained the same shallow logistic probes. Results:

Probe panel	Samples	Classes	Macro-F1	Balanced acc.
Real study, high-only	13,326	80	0.456	0.495
Real study, conservative-safe	4,751	26	0.518	0.537
Healthy vs Diseased	13,332	2	0.796	0.801

The dummy balanced accuracy for 80 studies is only 0.0125, so a balanced accuracy near 0.495 means the pretrained sample embeddings contain substantial real study information. The conservative-safe panel is not weaker; it is actually slightly easier to classify by study, probably because it contains fewer but more coherent studies.

14.4 Real study-id phenotype confounding

The real study labels confirm the central risk: study id and phenotype are not independent.

Panel	Studies	Cramer's V	NMI	Studies with top phenotype ≥ 0.8
High-only real study	80	0.792	0.526	54
Conservative-safe real study	26	0.634	0.528	0

For the full high-only real-study panel, 52 of 80 studies are phenotype-pure under the available labels. This means that blindly removing all real-study information would also remove disease-associated structure. The conservative-safe filter is therefore still important: it does not remove all confounding, but it avoids the most obvious single-phenotype studies.

14.5 Real study-id adversarial correction trials

We then reran GRL/DAB adversarial correction using true study ids. This implementation continues phase-2 masked-abundance training and adds a domain-adversarial batch classifier on the `<cls>` embedding. We also added warm-up parameters:

- `dab_warmup_epochs`: gradually increases DAB loss weight.
- `grl_warmup_epochs`: gradually increases gradient-reversal strength.

The exact loss remains:

```
total loss = masked-abundance reconstruction loss + effective_dab_weight * batch
CE
```

with gradient reversal applied to the batch classifier input, so the discriminator learns to predict study while the encoder receives the reversed gradient.

Trial	Study macro-F1 before	Study macro-F1 after	H/D before	H/D
Safe, DAB 0.05, batch 8, warm-up	0.518	0.580	0.796	
Safe, DAB 0.10, batch 8, warm-up	0.518	0.547	0.796	
High-only, DAB 0.05, batch 16, warm-up	0.456	0.493	0.796	

This is an important negative result. The model fine-tuning runs did not remove real study signal; independent post-hoc probes found study signal increased. At the same time, H/D performance stayed stable or improved. The most likely interpretation is that the reconstruction objective plus weak adversarial pressure continued to organize the embedding space around real cohort/phenotype structure, while the DAB objective was not strong or stable enough to force study-invariant representations.

14.6 Masked-abundance reconstruction after real-study GRL

Because BiomeGPT is fundamentally a masked-abundance foundation model, we also evaluated the main pretraining objective directly. We used the same deterministic nonzero-species mask for all checkpoints and computed masked abundance-bin MSE.

Checkpoint	Overall MSE	Safe-panel MSE	High-only MSE
Base phase-2	34.856	36.221	34.857
Safe GRL, DAB 0.05	33.235	33.852	33.235
Safe GRL, DAB 0.10	33.164	33.755	33.165
High-only GRL, DAB 0.05	32.911	34.225	32.911

So the GRL trials did preserve and even improve masked-abundance reconstruction. The failure mode is not catastrophic forgetting. The failure mode is more specific: the model remains, or becomes more, linearly study-separable in the `<cls>` embedding.

14.7 Representation-level real-study correction baseline

To test whether real study signal is removable at all, we implemented a transparent post-hoc embedding correction baseline:

```
x_corrected = x - cross_fitted_study_mean + cross_fitted_global_mean
```

This is a mean-only study correction. We also tested a stronger mean+scale version that maps each study's embedding variance to the global variance. The correction is cross-fitted with stratified folds, so each sample is corrected using study means estimated from other samples, not from itself.

Panel	Method	Study F1 before	Study F1 after	H/D F1 before	H/D F1 after
High-only	Mean-center	0.456	0.002	0.795	0.473
High-only	Mean+scale	0.456	0.008	0.795	0.471
Conservative-safe	Mean-center	0.518	0.004	0.690	0.637
Conservative-safe	Mean+scale	0.518	0.031	0.690	0.644

This result is very useful. It shows that study signal can be removed from embeddings when we explicitly subtract study-level structure. However, it also shows why broad study correction is dangerous: high-only correction nearly destroys H/D separability. The conservative-safe panel has a much smaller biological cost, which supports the earlier decision to avoid obviously phenotype-confounded study labels.

14.8 Updated interpretation from real study ids

The real-study-id experiments change the story from "we have a promising light GRL correction" to a more precise conclusion:

> Real study signal is strong and removable, but it is entangled with phenotype. Current model-level GRL fine-tuning preserves reconstruction but does not yet remove real study signal. Explicit embedding-level study centering removes study signal, and the biological damage is acceptable only under conservative-safe filtering.

The next model-level step should therefore not simply increase epochs. It should translate the successful representation-level behavior into the training objective, for example by adding a mean-alignment penalty such as study-centroid matching/CORAL/MMD on `<cls>` embeddings while keeping masked-abundance reconstruction as the primary loss, and selecting checkpoints by three criteria: lower real-study probe, preserved masked-abundance MSE, and preserved H/D probe.

14.9 Follow-up: representation alignment inside training

We then tested whether the successful post-hoc real-study centering could be moved into model training. Two model-level variants were implemented.

First, we added minibatch centroid alignment:

```
loss = reconstruction loss + alignment_weight * centroid_alignment_loss
```

The training loader was changed to use balanced study minibatches, so each minibatch contains multiple studies and multiple samples per study. This makes centroid estimates meaningful inside the batch.

Second, we implemented cross-fitted centroid distillation. We first compute the same corrected targets that worked in post-hoc analysis:

```
target = base_embedding - cross_fitted_study_mean + cross_fitted_global_mean
```

Then the model is trained with:

```
loss = reconstruction loss + target_weight * MSE(unmasked_<cls>, target)
```

An important debugging point: the target loss must be applied to the unmasked `sample_prompt` embedding, because that is the representation used by the probes. Applying it to the masked pretraining embedding does not directly constrain the evaluated embedding.

Method	Study F1 before	Study F1 after	H/D before	H/D after	Recon MSE
Base phase-2	0.518	–	0.796	–	34.856
Centroid align w10	0.518	0.598	0.796	0.807	33.697
Centroid align w50	0.518	0.579	0.796	0.819	33.628
Unmasked distill w10	0.518	0.532	0.796	0.801	33.656
Unmasked distill w50	0.518	0.556	0.796	0.802	33.587
Unmasked distill w200	0.518	0.576	0.796	0.800	33.656

These are useful negative results. All model-level alignment trials preserve or improve masked-abundance reconstruction, and H/D signal remains usable. However, none of these unconstrained backbone fine-tuning runs reduce real-study predictability. The independent study probe remains stronger than the original phase-2 embedding.

The practical result is therefore:

> Direct cross-fitted study centering currently works as a corrected representation, but simple GRL, minibatch centroid alignment, and centroid-target distillation are not yet sufficient to make the BiomeGPT backbone itself study-invariant.

The corrected embedding files were saved for downstream use:

- `dataset_v3/outputs_real_study_embedding_correction_saved/real_study_high_only_mean_center_corrected_embeddings.npz`
- `dataset_v3/outputs_real_study_embedding_correction_saved/real_study_conservative_safe_mean_center_corrected_embeddings.npz`

14.10 Follow-up: scGPT-style batch-conditioned decoder

Because scGPT uses batch-aware mechanisms such as batch labels, batch embeddings, and domain-specific batch normalization, we also tested whether BiomeGPT failed because it had no batch side-channel. The hypothesis was:

> If study id is available to the reconstruction decoder, reconstruction can model study-specific abundance shifts without forcing the final sample embedding to carry all study information.

We implemented a batch-conditioned residual reconstruction decoder:

```
pred = base_reconstruction_head(h_species) + residual_head(h_species, study_id)
```

The study id is not given to `sample_prompt`; it is only used by the residual decoder. The sample embedding is still trained toward the cross-fitted corrected target. We also tested an optional DAB loss and a detached reconstruction branch where reconstruction gradients do not update the encoder.

Trial	Study F1 before	Study F1 after	H/D before	H/D after
Batch decoder, normalized target w50	0.518	0.565	0.796	0.805
Batch decoder + DAB 0.1	0.518	0.577	0.796	0.804
Batch decoder, detached recon encoder	0.518	0.592	0.796	0.796
Batch decoder, raw target w1	0.518	0.558	0.796	0.803

This did not solve real-study removal. The trials preserved H/D signal, but the independent real-study probe remained stronger than the base phase-2 embedding. The important interpretation is that adding a decoder-side batch residual only during late continued training is probably not equivalent to scGPT-style batch-aware pretraining. The batch-aware mechanism likely needs to be part of the architecture and training dynamics from earlier in pretraining, or implemented as a frozen embedding correction adapter rather than as unconstrained backbone fine-tuning.

14.11 Follow-up: batch-token continued pretraining

We then created a cleaner batch-aware continued-pretraining script:

`dataset_v3/batch_token_pretraining.py`

During masked-abundance pretraining, the model sequence is:

```
[CLS, STUDY, species_1, species_2, ...]
```

The attention mask lets species tokens attend to the `STUDY` token, so reconstruction has a batch side-channel. However, `CLS` does not attend to `STUDY`, and downstream embeddings are still extracted with the original unconditioned `model.sample_prompt(bins)`.

Trial	Study F1 before	Study F1 after	H/D before	H/D after	Recon MS
Base phase-2	0.518	–	0.796	–	34.8
Batch token, reconstruction only	0.518	0.529	0.796	0.799	33.5
Batch token + DAB 0.1	0.518	0.548	0.796	0.805	33.5
Batch token + corrected-target distill	0.518	0.544	0.796	0.800	33.5

This is the most stable model-level batch-aware design so far. It improves masked-abundance reconstruction and keeps H/D signal, while the reconstruction-only run increases study separability only mildly. However, it still does not reduce real-study probe performance below the base phase-2 embedding. Adding DAB or corrected-target distillation made the independent study probe stronger.

The interpretation is that a true study token side-channel helps reconstruction, but adding it after the phase-2 representation is already formed is probably too late to make the final sample embedding batch-invariant. To fully test the scGPT-style idea, the batch-token mechanism should be present earlier in pretraining, ideally before or at the start of phase2.

14.12 Recommended experimental design

The next experiment should be a controlled conservative-safe hyperparameter sweep:

- Label panel: conservative-safe only.
- Batch size: `8` or `16`.
- DAB weight: `0.02`, `0.05`, `0.1`.
- GRL/DAB schedule: gradual warm-up over the first 30–50% of epochs.
- Model selection: lower conservative-safe batch probe, preserved H/D probe, and no large reconstruction degradation.

15 Files Produced

File	Description
dataset_v3/batch_effect_diagnostics.py	Script for extracting sample prompts, training batch/phenotype probes, computing confounding, and saving PCA plots.
dataset_v3/batch_adversarial_correction.py	Script for adversarial correction using safe, high-only, or high+medium label panels, including GRL/DAB warm-up options.
dataset_v3/prepare_real_study_batch_annotation.py	Script that parses the uploaded real study-id CSV, aligns it to phase-2 gut samples, and creates the real-study batch annotation table.
dataset_v3/real_study_embedding_correction.py	Cross-fitted real-study embedding correction baseline using mean-center or mean+scale correction.
dataset_v3/outputs_batch_diagnostics_taxonomy/batch_effect_diagnostics_summary.json	Main diagnostic summary.
dataset_v3/outputs_batch_diagnostics_real_study/batch_effect_diagnostics_summary.json	Real-study-id diagnostic summary using the uploaded sample-to-study mapping.
dataset_v3/outputs_batch_diagnostics_taxonomy/probe_metrics.csv	Batch, phenotype, and H/D probe metrics.
dataset_v3/outputs_real_study_embedding_correction/*compact_metrics.csv	Compact before/after metrics for cross-fitted real-study embedding correction.
dataset_v3/outputs_batch_reconstruction_diagnostics_real_study/reconstruction_diagnostics_summary.csv	Masked-abundance reconstruction MSE for the base and real-study GRL checkpoints.
dataset_v3/outputs_batch_diagnostics_taxonomy/*confounding_by_batch.csv	Per-batch phenotype-confounding tables.
dataset_v3/outputs_batch_adversarial_taxonomy/taxonomy_checkpoint_stage2_batch_adversarial_safe.pt	Best current conservative adversarial correction checkpoint.
dataset_v3/outputs_batch_diagnostics_taxonomy/batch_correction_status.md	Short status summary for quick reporting.
dataset_v3/outputs_batch_diagnostics_taxonomy/batch_adversarial_trial_comparison.csv	Summary table comparing label-panel ablations and stronger correction runs.

16 Current Conclusion

The taxonomy-aware BiomeGPT phase-2 embeddings contain batch/study information, so the batch-correction idea is motivated. However, most candidate batch labels are strongly phenotype-confounded, so correcting on all batch labels would be scientifically risky.

The earlier proxy-label conservative-safe adversarial trial provided a technical proof-of-concept: the full pipeline ran, and one light batch-size-8 run reduced safe-batch predictability with only a small drop in H/D signal. The newer real-study-id reimplementation is stricter and more informative. With true study ids, the current GRL fine-tuning preserves or improves masked-abundance reconstruction, but it does not remove real-study signal. Independent probes show real-study separability increases after the GRL trials.

The strongest useful result right now is the cross-fitted representation-level correction: study centering nearly eliminates real-study predictability. But the biological cost is large when all high-confidence real study labels are used, and much smaller when only conservative-safe studies are used. Therefore, batch correction is feasible and motivated, but final model-level correction should be conservative and should optimize against real study ids, masked-abundance MSE, and H/D preservation together.

17 Recommended Next Steps

1. Treat the uploaded real study ids as the main batch annotation going forward.
2. Use conservative-safe real-study labels first; broad high-only study correction is too phenotype-confounded.
3. Add a centroid-alignment/CORAL/MMD-style representation penalty to mimic the successful embedding-level study centering inside model training.
4. Select checkpoints by three criteria: lower real-study probe, preserved or improved masked-abundance MSE, and preserved H/D probe.
5. Fine-tune H/D from the best corrected checkpoint and compare ExVal metrics against the uncorrected taxonomy-aware checkpoint.